

Generation Interconnection System Impact Study Report

For Queue Project Q18-1130

Brent 138 kV
January 01, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1130. The Interconnection Customer proposes a 157.3 MW solar photovoltaic facility interconnecting at the Brent 138 kV substation in the Pennsylvania area. The request was received on December 31, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 452 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$204,593,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Dos Marinas 2003 / Brent 138 kV
Contingency: Southbase / Kings 138 kV
Loading: 180.5% (Rating: 275 MVA)
DFAX: 39.2%
Pre-project loading: 158.1%
2. Facility: Kings / Southbase 138 kV
Contingency: Brent / Dos Marinas 2003 Substation 138 kV
Loading: 158.8% (Rating: 331 MVA)
DFAX: 21.7%
Pre-project loading: 148.5%
3. Facility: Kings / Greenville Steam 138 kV L
Contingency: Clayton / Kings 138 kV
Loading: 152.8% (Rating: 190 MVA)
DFAX: 6.0%
Pre-project loading: 147.8%
4. Facility: Elk City Llc 345 kV/1 kV Transformer Bank
Contingency: Talenenergy Martins Crk Llc Harrisburg / Little Chute 345 kV
Loading: 149.3% (Rating: 487 MVA)
DFAX: 7.0%
Pre-project loading: 147.0%
5. Facility: Kings / Clayton 138 kV
Contingency: Greenville Steam / Kings 138 kV line
Loading: 148.3% (Rating: 228 MVA)
DFAX: 8.0%
Pre-project loading: 142.8%
6. Facility: Elk City Llc 138 kV/1 kV TX
Contingency: Talenenergy Martins Crk Llc Harrisburg / Little Chute 345 kV L
Loading: 143.7% (Rating: 506 MVA)

DFAX: 7.0%

Pre-project loading: 141.5%

7. Facility: Bemidji / Metalcan 138 kV

Contingency: Little Chute / Talenenergy Martins Crk Llc Harrisburg 345 kV L

Loading: 143.6% (Rating: 246 MVA)

DFAX: 7.7%

Pre-project loading: 138.7%

8. Facility: Union Power Station / Proctor 138 kV

Contingency: Dos Marinas 2003 345 kV/138 kV Auto

Loading: 136.4% (Rating: 202 MVA)

DFAX: 17.4%

Pre-project loading: 122.8%

9. Facility: Brent / Southbase 138 kV

Contingency: Dos Marinas 2003 / Brent 138 kV line

Loading: 133.8% (Rating: 191 MVA)

DFAX: 42.7%

Pre-project loading: 98.6%

10. Facility: Union Power Station / Clayton 138 kV line

Contingency: Dos Marinas 2003 345 kV/138 kV Transformer

Loading: 132.5% (Rating: 227 MVA)

DFAX: 17.4%

Pre-project loading: 120.4%

11. Facility: Hampton / Proctor 138 kV

Contingency: Dos Marinas 2003 345/138 kV Transformer

Loading: 132.2% (Rating: 192 MVA)

DFAX: 17.4%

Pre-project loading: 118.0%

12. Facility: Deerfield 5 / Hunterstown Power Plant 138 kV

Contingency: Hastings / Elk City Llc 345 kV

Loading: 128.5% (Rating: 261 MVA)

DFAX: 7.4%

Pre-project loading: 124.1%

13. Facility: Rainey / Hood Riv 138 kV

Contingency: Hastings / Elk City Llc 345 kV

Loading: 124.7% (Rating: 304 MVA)

DFAX: 7.4%

Pre-project loading: 120.9%

14. Facility: Rainey / Mtn 138 kV

Contingency: Hastings / Elk City Llc 345 kV

Loading: 123.9% (Rating: 312 MVA)

DFAX: 7.4%

Pre-project loading: 120.2%

15. Facility: Sawtelle / Hunterstown Power Plant 138 kV

Contingency: Hastings / Elk City Llc 345 kV

Loading: 123.8% (Rating: 285 MVA)

DFAX: 7.4%

Pre-project loading: 119.8%

16. Facility: Cardwell / Bemidji 138 kV

Contingency: Talenenergy Martins Creek Llc Harrisburg / Little Chute 345 kV Line

Loading: 121.2% (Rating: 303 MVA)

DFAX: 7.7%

Pre-project loading: 117.2%

17. Facility: Sawtelle / Hood Riv 138 kV line
Contingency: Elk City Llc / Hastings 345 kV
Loading: 115.8% (Rating: 312 MVA)
DFAX: 7.4%
Pre-project loading: 112.1%

18. Facility: Elk City Llc / John Hart 138 kV
Contingency: Elk City Llc / Little Chute 345 kV
Loading: 108.0% (Rating: 236 MVA)
DFAX: 6.6%
Pre-project loading: 103.6%

19. Facility: Freeport / Hampton 138 kV Line
Contingency: Dos Marinas 2003 345-138 kV XFMR
Loading: 105.9% (Rating: 220 MVA)
DFAX: 17.4%
Pre-project loading: 93.5%

20. Facility: Dos Marinas 2003 345/138kV Auto
Contingency: Orchard Road / Ft Humbug 500 kV
Loading: 104.6% (Rating: 343 MVA)
DFAX: 27.8%
Pre-project loading: 91.9%

21. Facility: Dos Marinas 2003 / Consumers Energy 138 kV line
Contingency: Dos Marinas 2003 345/138kV TX
Loading: 100.7% (Rating: 222 MVA)
DFAX: 12.7%
Pre-project loading: 91.8%

22. Facility: 12Th & Irving / John Hart Substation 138 kV Line
Contingency: Little Chute / Elk City Llc 345 kV Line
Loading: 100.2% (Rating: 219 MVA)
DFAX: 5.1%
Pre-project loading: 96.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dos Marinas 2003 / Brent 138 kV	Rebuild line with higher-capacity conductor	\$28,802,000	100% (\$28,802,000)
Kings / Southbase 138 kV	Rebuild line with higher-capacity conductor	\$12,502,000	80% (\$10,000,000)
Greenville Steam / Kings Substation 138 kV	Rebuild line with higher-capacity conductor	\$24,563,000	100% (\$24,563,000)
Elk City Llc 345/1 kV Transformer	Replace transformer with higher-rated unit	\$20,162,000	68% (\$13,699,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kings / Clayton 138 kV	Rebuild line with higher-capacity conductor	\$6,496,000	100% (\$6,496,000)
Elk City Llc 138/1 kV XFMR	Replace transformer with higher-rated unit	\$4,060,000	68% (\$2,759,000)
Bemidji / Metalcan 138 kV	Rebuild line with higher-capacity conductor	\$1,419,000	50% (\$710,000)
Union Power Station / Proctor 138 kV	Rebuild line with higher-capacity conductor	\$2,911,000	100% (\$2,911,000)
Southbase / Brent 138 kV L	Rebuild line with higher-capacity conductor	\$28,450,000	100% (\$28,450,000)
Union ower Station / Clayton 138 kV line	Rebuild line with higher-capacity conductor	\$2,205,000	100% (\$2,205,000)
Hampton / Proctor 138 kV	Rebuild line with higher-capacity conductor	\$5,752,000	100% (\$5,752,000)
Deerfield 5 / Hunterstown Power Plant 138 kV	Rebuild line with higher-capacity conductor	\$8,356,000	100% (\$8,356,000)
Rainey / Hood Riv 138 kV	Rebuild line with higher-capacity conductor	\$7,125,000	100% (\$7,125,000)
Rainey / Mtn 138 kV	Rebuild line with higher-capacity conductor	\$5,862,000	100% (\$5,862,000)
Sawtelle / Hunterstown Power Plant 138 kV	Rebuild line with higher-capacity conductor	\$6,438,000	100% (\$6,438,000)
Cardwell / Bemidji 138 kV	Rebuild line with higher-capacity conductor	\$32,129,000	50% (\$16,086,000)
Sawtelle / Hood Riv 138 kV line	Rebuild line with higher-capacity conductor	\$9,323,000	100% (\$9,323,000)
Elk City Llc / John Hart 138 kV	Reconductor line	\$184,000	100% (\$184,000)
Hampton / Freeport 138 kV	Reconductor line	\$1,058,000	100% (\$1,058,000)
Dos Marinas 2003 345/138 kV TX	Replace transformer with higher-rated unit	\$19,189,000	100% (\$19,189,000)
Dos Marinas 2003 / Consumers Energy 138 kV line	Reconductor line	\$1,428,000	100% (\$1,428,000)
12Th & Irving / John Hart Substation 138 kV Line	Reconductor line	\$3,197,000	100% (\$3,197,000)
	Total allocated to project		\$204,593,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.